

Immersive Chemistry Lab - IP2a Testing Transcripts

Apurva Tripathi - 49996081 | DECO2300/7230

The transcripts below are the raw testing evidence used for the evaluation. Minor punctuation/line-break changes have been made only for readability; the participant meaning has not been changed.

Participant 1

Researcher: Hi, how are you doing?

Participant 1: Hello, I'm good, how are you?

Researcher: Uh, I'm good too. So can I have your consent to record this audio?

Participant 1: Absolutely, okay. Go ahead.

Researcher: Uh, thank you. So here I have an updated prototype of an immersive chemistry lab. Uh, basically, I have a chemistry lab simulator in in VR headset. So imagine you're using a VR controller to perform chemistry experiment, and then we can explore what happens when we put 2 reactants together and what happens when a reaction happens - reaction occurs and how you can explore it on a molecular level. Uh, is that right?

Participant 1: Yeah.

Researcher: Yeah, so while you go through this whole experience of the chemistry lab, I would like you to say what you're thinking out loud. I'll be here to guide you through that like what are the controls and everything, but uh, I'll try to uh, not give instructions to best step and see how you navigate through this. Is that okay?

Participant 1: All right, yeah. Uh looks pretty cool. Should I go and use it?

Researcher: Uh, yeah, just a 2nd. I play it for you. Sure. Uh, and I will explain the control. So basically the square brackets are for this one is for the right controller and this one is for the left controller. And once you press the square brackets, after that, you can use WASD to move the controllers back and forth. And G is for grabbing. Is that okay?

Participant 1: Okay, sounds good.

Researcher: Uh, okay, so now we can, like, uh, you can proceed and uh, work with.

Participant 1: Okay, so I see 2 beakers. Okay, let me, let me put this reactant in the container. Oh, okay. It's pretty easy. Okay. Okay, let me put this, the one in the right, I'll pour this one in as well. Oh, okay. Oh, yeah, I see. The reaction happens as happened. Uh, yeah. Exciting.

Researcher: Yeah, so, uh, once the reaction happened, uh, there was a uh, molecular focus there. So what did you think that is for? What do you think that is?

Participant 1: I, I think that was like to denote that a reaction was happening.

Researcher: Yeah, yeah. Okay, you can proceed.

Participant 1: Okay, pretty cool. Okay, so now I see, okay, so build H₂, uh, water. Okay, so I have to connect the 2 hydrogen atoms to one of this matter. Okay. Pretty cool. Okay, I've connected the one on the right. Go ahead and connect the one on the left. Oh, nice. Yeah, I see the H₂ molecule. Nice.

Researcher: Yeah. And you can just play around with the atoms even more too, like, yeah, yeah.

Participant 1: I see, I can, I can, like, connect and disconnect them. That's nice. I also see how, like, the oxygen atom is different in appearance than the hydrogen one, so that's nice. Uh, yeah. I like it.

Researcher: Yeah. Yeah, so that was it for now. Is it okay if I ask a few questions regarding this whole experience?

Participant 1: Yeah, sure, please go ahead.

Researcher: Uh, yeah, so which interaction felt most natural to you while using this?

Participant 1: I think most interactions were pretty natural. I could grab the beakers, I can pour them in. Um, so that was, uh, pretty accurate. I liked it. Same with the atoms. I was very easily able to grab them and pull them around and as I brought the oxygen atom closer to the hydrogen atom, I saw them snap. So yeah, I think most of it were pretty natural.

Researcher: Okay. Which interaction was least obvious or you found it a bit confusing?

Participant 1: I think so far everything seemed very intuitive to me. I saw in the, in the molecule formation section, I saw, the instructions were right there in text. So it made it very clear what I'm supposed to do. Um, In the 1st one, um, where I was pouring in the reactants, I think, uh, maybe it would have helped to have a label for the reactants, but that's minor. Mostly it was pretty intuitive and pretty easy.

Researcher: Yeah, okay, that makes sense. Did you feel like you could use both the controllers to perform the experiments or interact with the atoms?

Participant 1: Yes, I, uh, in the, in the, uh, reaction one, the 1st section, I use the right one to move the right container and the left one to move the left one and it possibly pretty simple.

Researcher: Okay, and could you use, like, even if you interchange the controller once you did that, you were able to use, use it for both the, I mean, for both the reactants, right? No problems there.

Participant 1: Yeah.

Researcher: Okay. And what did the arrows and instructions like there were a few instructions like arrows when the reaction happened or when you were in the molecular phase. were they clear or were they confusing?

Participant 1: No, they were pretty clear.

Researcher: Okay. Okay. Oh, what is one change you would like to make for the next prototype if there's any suggestion feedback?

Participant 1: I guess, just that I would have the labels for the reactants, um, like, okay, maybe there's a copper sulphate and maybe I'm pouring an iron or whatever. Okay. But beyond that, I think it looks pretty good to me.

Researcher: Okay, okay. Thank you so much. Thank you for your feedback.

Participant 1: Thank you.

Participant 2

Researcher: Hi, how are you doing?

Participant 2: Hi, I'm fine, how are you?

Researcher: I'm good to, can I have your consent to record this audio?

Participant 2: Yeah, yeah sure.

Researcher: Okay, thank you. Uh, so basically this is an, uh, updated prototype of, uh, of immersive chemistry lab that I'm working on. This is a VR simulator for a chemistry lab. So for this, I would like you to imagine that you are using a VR controller to perform chemistry experiments, and then we can explore what happens when once the reaction happens and how you can interact with molecules in that reaction. Uh, so is that clear?

Participant 2: Yeah, it's okay.

Researcher: Okay. So, uh, also I will be, uh, I will be telling you what the controls are and I'll be here to, um, guide you through the whole process, but I would like you to think out loud while you're using the, uh, using the application, if that's okay.

Participant 2: Okay, okay, got it.

Researcher: Okay, thank you. Okay, so now you can like start using it. Also, sorry, before I'll just tell you what the controls are. Um, just a 2nd, get it play, and then I'll uh, I'll explain the controls. I think it's pain now. Yeah. Okay. So basically the square brackets are for the controller. The right one is for the right controller and left one is for left controller. And once you click on the square brackets, after that, you can use WASD to move the controllers forward and backward, and you can use G to grab any object. Is that okay?

Participant 2: Yeah, yeah, got it.

Researcher: Okay, so now you can start using the controller and uh, I would like you to like think out loud and just, you know, let me know what you are going when you're using the whole uh, application.

Participant 2: Alright. So, um, moving the control right now. Okay. Uh, so I have to pour the things into the weaker, right?

Researcher: Yeah, yeah.

Participant 2: Okay. I'm gonna do that now. Okay. Okay, so it's saying expand your hands to explore molecular scales. So I have to expand the controllers, right?

Researcher: Yeah, yeah, you're right.

Participant 2: I'm gonna do that, no? Right.

Researcher: Yeah, so what do you think happened once you expanded the controller?

Participant 2: So I think, uh, I, I can see the molecular reaction of what, uh, like the, Like the molecules, right? Uh, I think there can be a new like uh, animation or something that takes me to the molecular reaction. Do you get what I'm saying?

Researcher: Yeah, yeah, right.

Participant 2: Like in between these 2 screens that and like uh, microanimation that shows that I'm now zooming into the molecules. basically.

Researcher: Mm-hmm. Okay.

Participant 2: Now, it's saying connect to hydrogen atoms to one oxygen atoms. Okay. I'm gonna connect it. Okay and a water molecule is formed. So it turns blue, the when it gets connected, the molecules turn blue. Is that what's happening?

Researcher: Oh, when it's in the grabable range, like, uh, it tells you that, okay, yeah, now you can grab this.

Participant 2: Okay, okay, got it. So once, uh, like a water molecule has been formed, there can be another, like, uh, notable difference, like maybe the colour changes or something like that, which shows that now there's like a form, combination is being formed basically. That can happen, which will, like, you know, using it. There should be a. There should be like a surprise, like surprising new interaction which will happen in the final stage of the combination, basically is what I'm saying.

Researcher: Okay, okay. Yeah, that makes sense. Okay, so here the whole interaction ends. Okay, I'll pause it. Uh, can I ask a few questions regarding the whole experience?

Participant 2: Yeah, sure.

Researcher: Okay, so which interaction felt the most natural to you?

Participant 2: I think the whole flow was natural. Uh, it obviously needs a bit of uh, more like animated uh, elements to make it more fun. But I think the rest of it is natural.

Researcher: Okay. Okay. And was there an interaction that felt confusing to you?

Participant 2: No, no, it was clear. like there were steps and uh, different like, you added some text, right, to that they follow, basically. So I like that, I was not confused at all throughout the flow. I think there can be more gamification to make it more attractive. But uh, except for that, I think it was good.

Researcher: Okay. Uh, did you feel like uh, you could use both the controllers to perform uh, like basically the interactions that were happening?

Participant 2: Yeah, the controls were smooth and easy to use as well.

Researcher: Okay. And just one last question that I would like to ask is, is there any change that you would like to see in the next prototype or any feedback suggestion?

Participant 2: Yeah, I've given the feedback, I think, uh, the changes, uh, I've mentioned already, like, you can add in interactive, like, uh, animation or gamified animation or something, in, like, uh, when a combination is formed, you can add something that shows, like, there's like a different, uh, Uh, like new element is being formed

with, okay? Except for that, I think it was good and easy to use. It was, uh, it's a good informative, uh, like uh, experience. Uh, yeah, that's it.

Researcher: Okay, and sorry, just one more question. I wanted to ask, is, like, are the labels clear? Were they able to inform you what is happening or what you need to do next?

Participant 2: Yeah, yeah, I think it was clear.

Researcher: Okay. Okay, so here, like, this is the whole experience, and that was it, yeah. Thank you.

Participant 2: Okay, thank you so much. Thank you.